Low-Flow Turbine: Design Summary (current design)

8-inch hydrokinetic turbine for 0.5-1.0 m/s streams, CubeMars GL60 KV25, v5 simplified prototype with the v4 optimised blade. Model predictions only; nothing has been built or measured. Generated by `make_summary_v5.py` from the result files.

1. Current design and predicted output

Rotor	3 blades, NACA 4412 to 4418 root, design TSR 1.5, pitch -3 deg, chord 50 / 60 / 31 mm (hub / max / tip), Ø203.2 mm, hub Ø56
Assembly	one-piece printed rotor, printed housing (one 12 mm lip seal + one 6001 bearing), printed mount = rear lid + stake socket, small bell adapter; GL60 dry inside
Net 12 V output at 0.5 / 0.7 / 1.0 / 2.0 m/s	0.14 / 0.93 / 3.41 / 27.2 W
Self-start / sustain	0.6 / 0.5 m/s (hand spin covers 0.5 m/s)
Operating speed, current, bus voltage at 0.7 m/s	184 rpm, 0.16 A, 7.0 V
Open-circuit at 2 m/s	46 V (below the 60 V limit)
Depth	91 mm nose to lid; packs as rotor 59 mm + housing/mount 76 mm

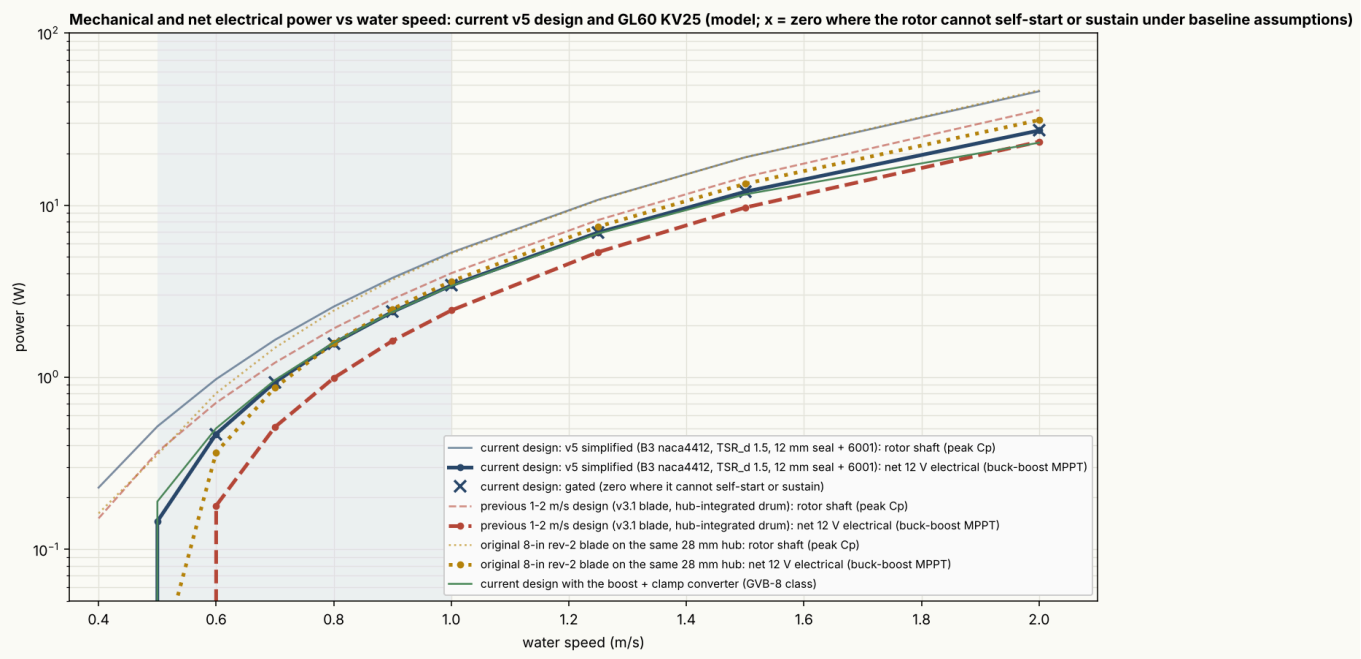


Figure 1. Mechanical and net electrical power against water speed for the current design, the original rev-2 blade and the v3.1 drum design. Crosses: zero where the current design cannot self-start or sustain.

2. Motor: listed data and how we run it

CubeMars GL60 KV25 (listed)	**value**
rated voltage / power	24 V / 20 W
rated torque / current / speed	0.6 N m / 1.35 A / 310 rpm (one reseller: 350 rpm / 1.43 A)
peak torque / current	1.75 N m / 4 A

CubeMars GL60 KV25 (listed)	value
no-load speed	516 rpm at 24 V → KV 21.5 rpm/V (we use this; the "KV25" name is a different convention)
resistance / inductance	5.5 ohm / 2.72 mH
pole pairs, mass, size	14, 230 g, 69 × 22.3 mm
bore, rating	20 mm hollow shaft (most listings; the encoder variant is listed with 6 mm), IP45 per reseller listings only (not submersible either way: the housing keeps it dry)

Source: the CubeMars specification table as returned by web search (cubemars.com product and goods pages; reseller listings agree except one outlier quoting 1140 rpm). The manufacturer page could not be fetched directly from this session. No current-speed curve is published; Figure 2 draws the lines the model uses from the listed constants and marks where the turbine runs the motor. The generator is used far below its ratings: 0.36 A at 1 m/s, 1.26 A at 2 m/s against 1.35 A rated; the 1.75 N m peak is a motoring figure, not a generating one.

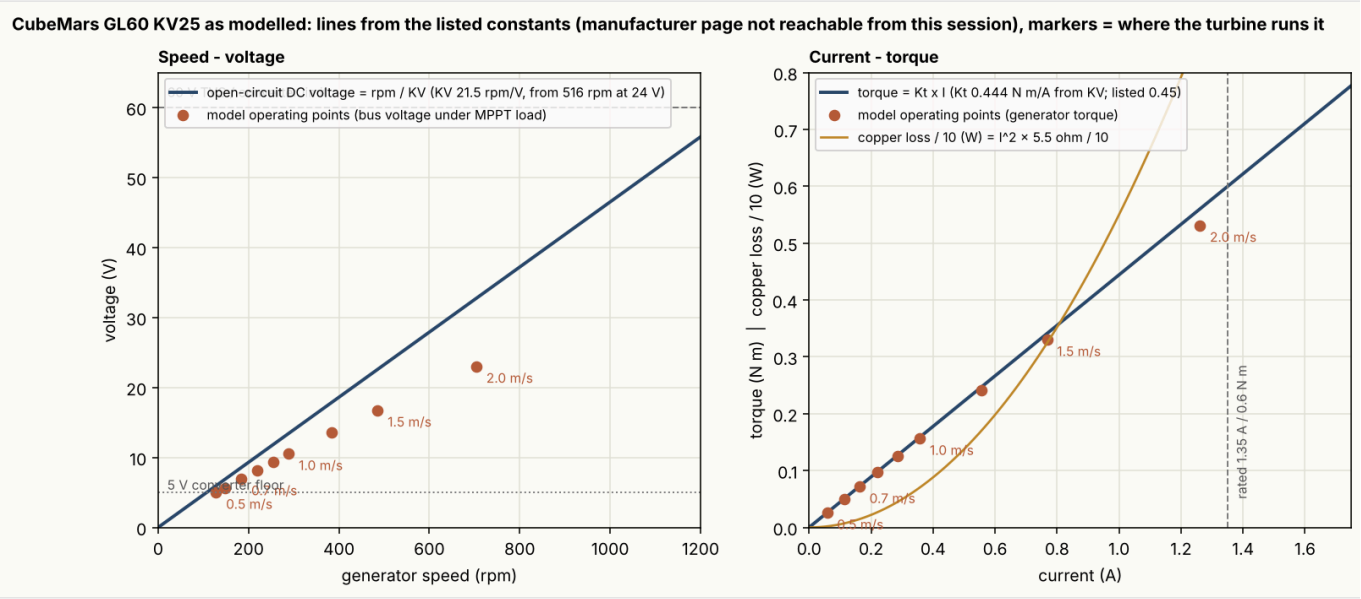


Figure 2. GL60 KV25 as modelled: open-circuit voltage = rpm / KV, torque = Kt x I with Kt = 0.444 N m/A, copper loss I^2 R; markers are the model's operating points from 0.5 to 2 m/s.

3. Assumptions made during development

- Water 1000 kg/m3, 1.0e-6 m2/s; uniform inflow; a 15 % inflow deficit inboard of r 38 mm behind the housing (reviewer's estimate 0.6-0.75, carried as a sensitivity).
- Airfoil polars from NeuralFoil (XFOIL surrogate), not measurements; wet printed-surface transition unknown (Re x/÷1.5 sensitivity).
- Generator: KV 21.5 from the listed no-load speed, R 5.5 ohm, cogging 24 mN m (assumed; the single largest unknown for self-start), no-load loss 0.5 W at 700 rpm, hole patterns PCD 40 (bell) / 50 (base), 20 mm bore.
- Friction: 12 mm PTFE lip seal 6 mN m running / 12 breakaway, 6001 bearing 3 / 4 mN m.
- Electronics: Schottky bridge (0.30 V + 0.20 ohm per diode), buck-boost MPPT with 94 % peak efficiency, 0.13 W fixed overhead, 5 V input floor, never more than 1.35 A drawn.
- Starting torque = BEM torque at TSR 0.1 with induction (the no-induction estimate, 1.8x higher, is an upper bound only).
- Printed ASA allowables 8 MPa long-term wet; thrust goes through the GL60 bearings in v5 (axial play to be checked).

4. The HW7 spreadsheet: how it worked and what it did

The sheet is a 20-strip Betz-optimum blade calculator: design TSR, Schmitz chord 8 pi r (1 - cos phi) / (B CL), twist = phi - alpha, closed-form induction, Prandtl tip and hub losses, torque summed over the strips, then a matched-load generator check (E = Ke x omega, P = (E - 2 Vf)^2 / 4R) against a required power and a minimum bus voltage. With the v4 inputs (sheet "v4 design" of `IED_HW7_v4_parameters.xlsx`) it gives Cp 0.30 and a chord within 2.3 mm and a twist within 1.1 deg of the optimised blade over the span (the tip taper is the only difference).

What it fixed in the outcome, and what it could not:

the sheet decided	the sheet could not see, so the workflow added
the design method itself: Schmitz planform + best-L/D twist at a design TSR	Reynolds-dependent polars for 7 sections (one CL/CD point in the sheet)
that the generator voltage, not the rotor, is the low-speed limiter: at the design TSR the sheet reports E = 4.6 V < 5 V floor and FAIL	the operating point from the generator load (TSR 2.8, 6.9 V at 0.7 m/s) instead of the design TSR
the pass/fail logic: power, torque and voltage checks against the motor	self-start gating against cogging + breakaway, which removed every design TSR >= 3
	iterated induction per element, Buhl correction, 508 candidates x 7 speeds, two energy objectives, robustness and sensitivity studies

The sheet was not an input file to the optimisation; it is the one-page hand version of stages 2-3 of the workflow and the cross-check of the result.

5. How we iterated

1. **Original 8-in rev-2 blade** (1-2 m/s design, TSR 4): 0.02 / 0.87 / 3.61 W at 0.5 / 0.7 / 1.0 m/s on the same hub, self-start 0.7 m/s.
2. **v3.1 drum design** (motor inside a 48 mm hub): 0.00 / 0.51 / 2.44 W, self-start 1.0 m/s; the same blade in the 48 mm drum delivers 30-44 % less at 1.0-0.7 m/s, so the small hub ahead of the housing was kept.
3. **v4 low-flow optimisation**: 384 stage-1 + 124 stage-2 candidates, two objectives, self-start gate, packaging rule; adversarial review (three reviewers, 47 findings) changed the starting-torque model and added five sensitivities; the search was re-run and the pick moved from design TSR 2.0 x1.3 to TSR 1.5 x1.0.
4. **v5 simplification**: same blade; shaft, collars, circlips, clamp ring and machined caps replaced by three printed parts and one adapter; cost 3 % at 0.7 m/s for the 12 mm seal.
5. **Diameter as an iteration variable** (Figure 3, same blade scaled, hub and motor fixed): smaller is worse than the area ratio because chord Reynolds number, hub ratio and fixed losses all move the wrong way, and the starting torque falls as D^3 against a fixed 40 mN m breakaway (cogging + seal + bearing).

diameter	net W at 0.5 / 0.7 / 1.0 m/s	self-start (m/s)	Cp peak at 1 m/s	starting torque at 0.5 m/s (mN m)	thrust at 2 m/s (N)
8 in	0.14 / 0.93 / 3.41	0.6	0.327	29 vs 40	38
7 in	0.00 / 0.48 / 2.14	0.8	0.301	18 vs 40	30
6 in	0.00 / 0.14 / 1.08	1.25	0.266	10 vs 40	22
5 in	0.00 / 0.00 / 0.32	1.5	0.211	5 vs 40	15

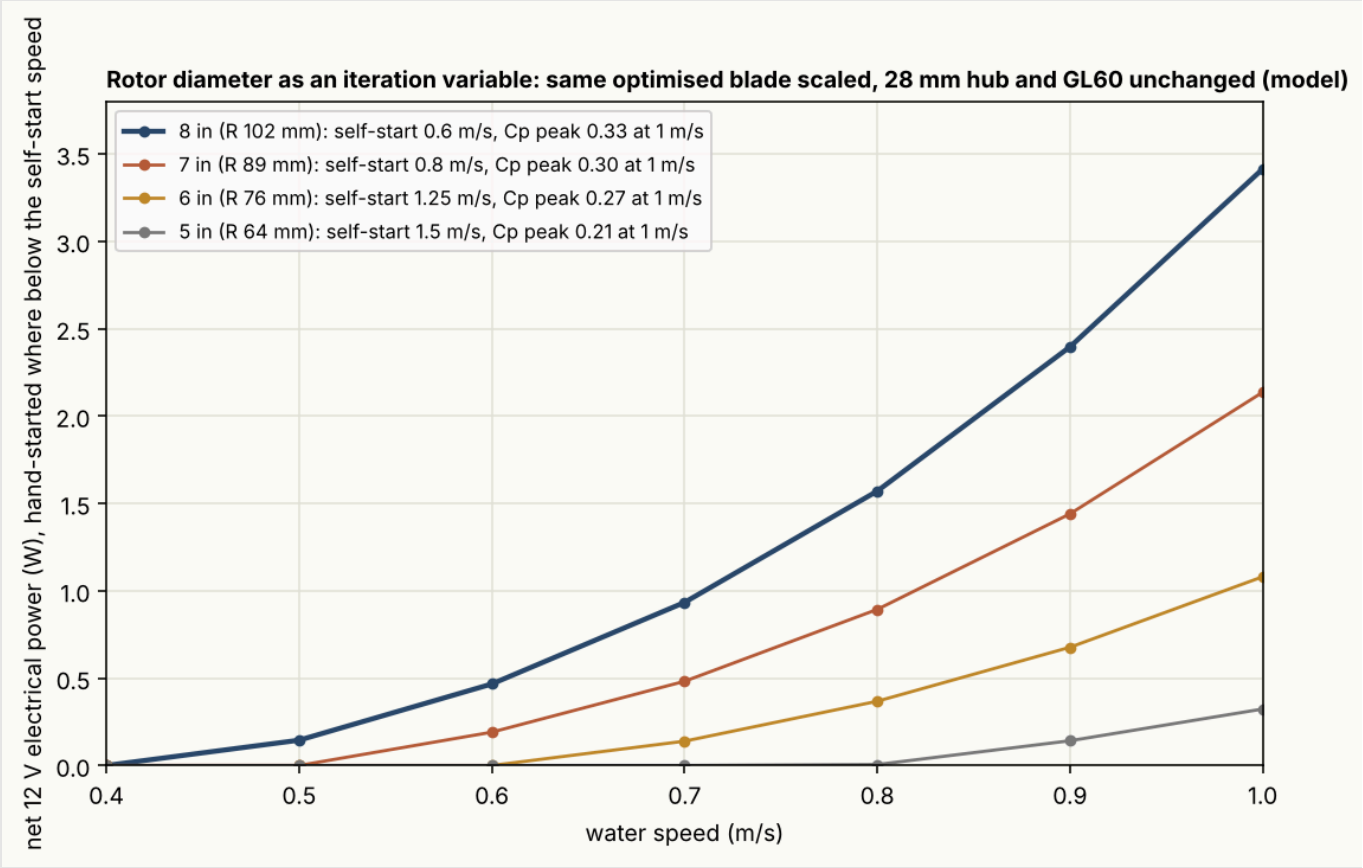


Figure 3. Rotor diameter study. 8 in is the smallest diameter that produces anything useful at the low end with this generator.

6. Foil, blade count, final pick

- **Foil:** NACA 4412 (L/D 10 / 49 / 79 at Re 4e4 / 8e4 / 2e5) beat the higher-lift SG6043 and FX 63-137 in the coupled result (1.51 vs 1.36 and 1.34 W mean net power): their higher design lift coefficient gives narrower chords, lower Reynolds number and a sharper Cp peak. E818 and E387 were excluded for low surrogate confidence. NACA 4418 at the root for printability.
- **Blade count:** best 2 / 3 / 4 / 5-blade candidates score 1.40 / 1.51 / 1.51 / 1.47 W and start at 0.7 / 0.6 / 0.6 / 0.5 m/s: two blades start late, five lose Cp to drag; three is the compromise within a few percent of four.
- **Final pick:** the highest mean of the two normalised objectives among candidates with a printable hub wall; it self-starts from 0.6 m/s (29 vs 40 mN m at 0.5 m/s in the v5 arrangement), has the highest output at 0.5-0.7 m/s once turning, and stays within 5 % of the original blade at 0.8-1.0 m/s. The eight best candidates were within 1.5 % of each other.

7. Assembly check (current CAD)

Assembly STEP `out/v5_assembly.step`: 32 parts; all printed parts valid single solids: True; all parts single solids: True; rigid-part clash check over 465 pairs: none; seal lip overlap as modelled: True; STLs watertight: True. The NX archive supplied (`updated_nx_parts.zip`) holds the same 32 part names plus the assembly as Siemens NX `.prt` conversions of these STEP files; NX geometry cannot be read in this environment, so the check above is on the STEP source.

8. What to measure first

GL60 cogging torque and the DC voltage per rpm through a bridge; seal and bearing friction; rotor starting torque at 0.5-0.7 m/s with the shaft locked; the power curve. These replace the four assumptions that set the 0.5-0.7 m/s numbers.